Bio-Inspired Low-Cost Quadruped Robot for Research and Education

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***Abstract*—Quadruped robots have demonstrated significant potential for applications such as industrial inspection, infrastructure monitoring, and research in unstructured environments. However, the high cost of commercial platforms limits their adoption, particularly in educational and research institutions with restricted budgets. This paper presents the design, implementation, and experimental validation of a low-cost quadruped robotic platform developed using commercially available components, additive manufacturing techniques, and open-source software tools. The proposed architecture integrates the Robot Operating System 2 as the high-level robotic framework, Controller Area Network based communication for distributed control, and Dynamixel actuators for locomotion. Forward and inverse kinematic models were implemented together with a parametric trajectory generation strategy to enable legged locomotion. The platform was manufactured with a total cost of approximately USD 800 and a total weight of 2.4 kg and dimensions of $10 \times 20 \times 22$ cm . Experimental evaluations included structural analysis, actuator communication and positioning tests, robot stability assessment, and locomotion accuracy experiments. The results demonstrated reliable system operation, with actuator positioning errors below 2% and a locomotion positioning error of 9.06% over a 50 cm displacement task.**

***Index Terms*—Quadruped Robot, Bio-Inspired Robotics, Robot Operating System 2, Controller Area Network Communication, Low-Cost Robotic Platforms.**

I. Introduction

Quadruped robots have attracted significant attention in recent years due to their ability to traverse complex and unstructured environments that are often inaccessible to wheeled or tracked platforms [1]. Their versatility has enabled several applications such as industrial inspection [2], infrastructure monitoring [3], or disaster response [4]. Notable examples include the quadruped robots developed by ANYbotics and Boston Dynamics, namely ANYmal [5] and Spot [6]. These platforms are capable of autonomous navigation, obstacle avoidance, and advanced perception tasks, making them valuable tools for Industry 4.0 applications.

Despite their technological maturity, the widespread adoption of commercial quadruped robots remains limited due to their high acquisition costs. The purchase price of platforms such as ANYmal and Spot is typically beyond the financial capabilities of many educational institutions, particularly in developing countries, restricting their use to well-funded universities, research centers, and large industrial organizations. Consequently, access to advanced quadruped robotics research and education remains highly unequal across different regions of the world.

To address this challenge, several researchers have proposed low-cost quadruped platforms based on open-source

hardware, additive manufacturing, and commercially available components. Examples include Stanford Pupper [7], PADWQ [8], Iron Dog Mini [9] and Mini cheetah [10], which seek to reduce manufacturing costs while maintaining acceptable locomotion performance. However, even among these low-cost alternatives, the reported costs remain relatively high, with the most affordable platforms still requiring investments close to USD 2,000 or more. Such costs continue to represent a significant barrier for many universities in the Latin American context, where research budgets and laboratory resources are often limited.

Furthermore, some of the existing low-cost designs present additional technical limitations. For instance, the Iron Dog Mini platform proposed by Rahman *et al.* achieves a manufacturing cost of approximately USD 333 through the use of low-cost servo actuators and 3D-printed components [9]. However, in the reported open-source resources appear to be mainly focused on the mechanical design files, such as STL components. The paper does not explicitly report the availability of source code for locomotion control, simulation, actuator testing, or communication implementation. This limits the reproducibility of the complete robotic system, since reproducing the platform requires not only manufacturing the mechanical structure, but also implementing and validating the software, communication, and actuator-control layers.

Motivated by these challenges, this paper presents the development of a low-cost and replicable quadruped robot for research and education in Latin America. Developed in Colombia using commercially available components and additive manufacturing techniques, the platform has a total cost of approximately USD 800. Its development followed an adapted Rational Unified Process (RUP) methodology, documenting the requirements, design, implementation, and experimental validation stages. The open-source repository (GitHub/repository) provides step-by-step documentation, STL files, ROS 2 packages, embedded control code, communication implementation, and experimental resources [11]. This structured methodology allows students to explore and reproduce the robotic development process while providing researchers with a modular and accessible platform for further experimentation.

II. Theoretical Background

A. Mobile and bio-inspired Robots

Mobile robots are autonomous or semi-autonomous systems capable of moving through and interacting with their surrounding environment. Depending on the operating medium, mobile robots can be classified into different categories, including unmanned aerial vehicles (UAVs), which operate in aerial environments, and unmanned ground vehicles (UGVs), which operate on terrestrial surfaces [12]. Within the field of ground robotics, locomotion can be achieved through various mechanisms such as wheels, tracks, and legs. Wheeled and tracked robots offer advantages in terms of mechanical simplicity, energy efficiency, and ease of control; however, their mobility is often limited when traversing highly irregular, unstructured, or obstacle-rich environments. To overcome these limitations, legged robots have been developed to provide enhanced adaptability, maneuverability, and terrain negotiation capabilities. Consequently, legged robotic platforms have attracted significant interest for applications including military operations, logistics and transportation, environmental exploration, infrastructure inspection, and emergency rescue missions [13].

The development of legged robots is closely related to the field of bio-inspired robotics, which seeks to extract and adapt functional principles observed in biological systems to solve engineering challenges. Rather than directly replicating natural organisms, bio-inspired robotics abstracts relevant characteristics such as locomotion strategies, body morphology, sensory integration, and adaptive behaviors, and incorporates them into robotic designs. In particular, terrestrial animals have evolved highly efficient locomotion mechanisms that enable stable and agile movement across complex terrains. These biological capabilities have motivated the design of quadruped robots, whose mechanical structures and gait patterns are often inspired by four-legged animals. As a result, bio-inspired quadruped robots have become important research platforms for investigating mechanical design, kinematic and dynamic modeling, trajectory generation, gait planning, and locomotion control in challenging environments [14].

B. ROS 2 and CAN Protocol

The ROS 2 is one of the most widely adopted middleware frameworks in robotics research, education, and industrial applications [15]. ROS 2 provides a collection of software libraries, tools, and communication interfaces that facilitate the development of robotic systems. Its distributed architecture enables seamless integration between sensing, control, planning, and actuation modules, allowing efficient communication between software components and hardware devices [16].

The CAN protocol, developed by Robert Bosch GmbH in the early 1980s, is a multi-master communication protocol widely used in embedded and robotic systems. CAN operates as a broadcast network in which all nodes connected to the bus can receive transmitted messages. Each message contains an identifier that determines both its priority during arbitration and its relevance to individual nodes. The protocol supports communication rates of up to 1 Mbit/s and provides high reliability through differential signaling, error detection mechanisms, and automatic message retransmission. During bus arbitration, dominant bits (‘0’) override recessive bits (‘1’), ensuring that the highest-priority message gains access to the communication channel without data corruption [17].

III. Design and Validation Methodology

A. Development Procedure

The development of the proposed quadruped robotic platform followed a systematic engineering approach consisting of four main stages: requirements definition, system design, implementation, and experimental evaluation. Initially, the functional and non-functional requirements of the platform were identified using a Quality Function Deployment (QFD)

matrix to establish the relationship between user needs and engineering specifications [18]. The functional requirements included quadruped locomotion, remote operation, communication between the embedded control layer and the high-level processing unit, and interaction between the physical robot and the ROS 2 visualization environment. The non-functional requirements focused on low-cost implementation, modularity, open-source compatibility, distributed communication, embedded control, and battery-powered operation. Based on the identified requirements, the hardware and software architectures were defined to integrate the mechanical, electronic, communication, and high-level control subsystems. Fig. 1. illustrates the overall design and subsystem integration.

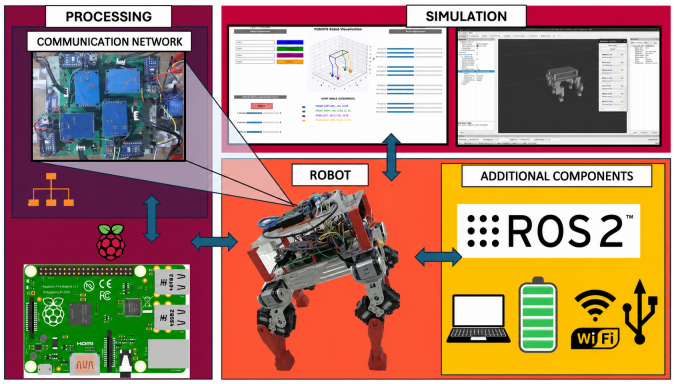


Fig. 1. Conceptual design of the proposed quadruped robotic platform, showing the integration between the mechanical structure, embedded control system, communication architecture, and ROS 2-based visualization environment.

B. Hardware and Software Architecture

The proposed quadruped robot uses a distributed control architecture that separates ROS2-based locomotion computation from embedded actuator control, improving modularity and facilitating future expansion. Kinematics run on a Raspberry Pi, which sends joint-angle references through a serial interface to an Arduino gateway acting as master of a CAN bus built with MCP2515 modules. The message dictionary, designed by the authors, follows a master–slave scheme: the master sends JSON messages whose identifier field indicates the destination leg or joint, followed by the data {"id":2,"q":[q1,q2,q3]} ; each node processes only its messages and replies in the same format. JSON was chosen because it is human-readable and easy to debug and extend, which helps teaching. Each node commands Dynamixel AX-12A+ servomotors, which provide position and velocity feedback at levels suitable for a low-cost platform, through a half-duplex bus with a 74LS241 buffer. Software was developed in Python3.10 with ROS2 Humble under Ubuntu22.04, and firmware in the Arduino IDE. The main hardware and software components employed in the proposed platform are summarized in Table I.

C. Kinematic Modeling

The locomotion of the proposed quadruped robot was achieved through the kinematic modeling of each robotic leg. The kinematic model establishes the relationship between the joint variables and the position of the foot end-effector, providing the mathematical foundation required for trajectory planning and motion control. Each leg was modeled as a serial kinematic chain composed of three rotational joints corresponding to the hip, femur, and tibia segments.

Forward kinematics was formulated using the Denavit–Hartenberg (D–H) convention, where homogeneous transformation matrices were employed to describe the position and orientation of the foot with respect to the leg reference frame [12]. This model was used to characterize the workspace of the leg and to verify the feasibility of the desired foot trajectories. To generate the joint commands required for locomotion, the inverse kinematic problem was solved using an iterative method based on the Jacobian transpose approach [19]. Given a desired foot position in Cartesian space, the joint configuration was updated iteratively according to

$$\theta_{k+1} = \theta_k + \alpha J^T(\theta_k)(x_d - f(\theta_k)), \tag{1}$$

where θ_k represents the current joint configuration, α is the step size, $J^T(\theta_k)$ denotes the transpose of the Jacobian matrix, x_d is the desired end-effector position, and $f(\theta_k)$ corresponds to the current end-effector position. This formulation allows the Cartesian trajectories generated for the foot to be converted into joint-space commands that can be executed by the actuators during locomotion.

TABLE I
MAIN HARDWARE AND SOFTWARE COMPONENTS OF THE PROPOSED PLATFORM

Category	Components	Cost (USD)
Hardware	Dynamixel AX-12A++ actuators	600.00
	Arduino Nano	40.00
	MCP2515 CAN modules	12.00
	74LS241 buffer	4.00
	5 V/10 A and 12 V/10 A power modules	15.00
	11.1 V 4000 mAh LiPo battery	30.00
	Others	100.00
Software	Arduino IDE 2.2.1	Free
	ROS 2 Humble	Free
	Python 3.10	Free
	Ubuntu 22.04	Free

D. Trajectory Generation

Foot trajectory generation plays a fundamental role in legged locomotion, as it defines the desired motion of the foot end-effector during the swing phase and directly influences gait characteristics such as stride length, foot clearance, and walking stability [20]. In the proposed platform, locomotion was achieved by generating a parametric Cartesian trajectory for the foot end-effector, which was subsequently converted into joint-space commands through the inverse kinematic model described in the previous subsection.

The adopted trajectory was designed to produce a smooth forward displacement while lifting the foot to avoid contact with the ground during the swing phase. As illustrated in Fig. 2, the trajectory is parameterized by the stride length a,

the lifting height h, and the nominal leg height z_0. These parameters allow the adjustment of the robot's locomotion characteristics according to the desired gait configuration.

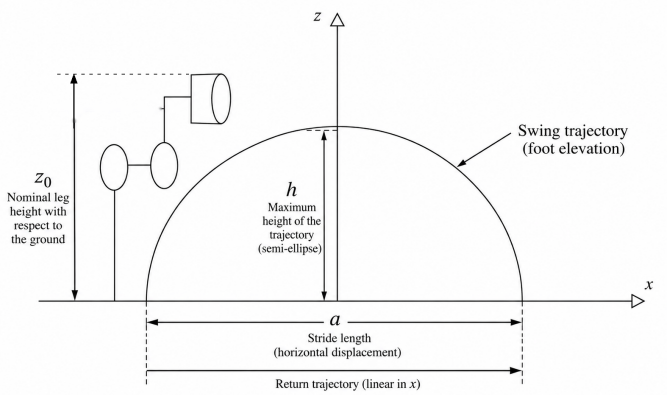


Fig. 2. Schematic representation of the robotic foot trajectory in the Cartesian x–z plane. The upper curve illustrates the sinusoidal swing motion, while the lower horizontal arrow indicates the linear return motion. Parameters a, h, and z_0 denote the stride length, maximum foot elevation, and nominal leg height, respectively.

The horizontal and vertical components of the trajectory are defined as:

$$x(t) = a\left(\frac{t}{N}\right), \tag{2}$$

$$z(t) = z_0 + h\sin\left(\pi\frac{t}{N}\right), \tag{3}$$

where t denotes the discrete trajectory point index and N is the total number of points used to complete one swing trajectory. The ratio t/N normalizes the trajectory progression from 0 to 1. In the experimental implementation, $N = 15$ was used. The horizontal coordinate $x(t)$ increases linearly from 0 to the stride length a, while the vertical coordinate $z(t)$ follows a sinusoidal profile: it starts at the nominal height z_0, reaches its maximum value $z_0 + h$ at the middle of the swing phase, and returns to z_0 when $t = N$. Different values of a and h were evaluated to analyze their influence on the reachable workspace and the feasibility of the generated locomotion trajectories.

IV. Results and Discussion

A. Prototype Implementation

The proposed quadruped robotic platform was successfully manufactured and assembled as a low-cost and modular robotic system. The final prototype consists of a rigid central body and four articulated legs, each incorporating three actuated joints corresponding to the hip, femur, and tibia. The robot was fabricated using additive manufacturing techniques and commercially available electronic components, resulting in a platform that can be easily reproduced for educational and research purposes. Figure 4 presents the final physical implementation of the proposed quadruped robot.

Prior to fabrication, a finite element analysis (FEA) was conducted to evaluate the structural behavior of the robot under loading conditions and to support the material selection process. Four materials commonly used in additive manufacturing were analyzed: polylactic acid (PLA), acrylonitrile butadiene styrene (ABS), polyethylene terephthalate glycol (PET-G), and polymethyl methacrylate (PMMA). The results are summarized in Table II.

TABLE II
Material comparison based on finite element analysis

Material	Maximum von Mises stress (MPa)	Safety factor
PLA	0.5358	12.0
ABS	0.9812	0.76
PET-G	1.258	9.3
PMMA	1.480	12.0

The analysis revealed that ABS exhibited the lowest structural performance, achieving a safety factor of only 0.76, which indicates a higher susceptibility to failure under the evaluated loading conditions. In contrast, PLA and PMMA achieved safety factors of 12, while PET-G presented a safety factor of 9.3. Although PLA exhibited a slightly higher safety factor, PET-G was selected for the fabrication of the prototype. The complete robot achieved a total mass of approximately 2.4 kg, including the mechanical structure, actuators, embedded electronics, communication modules, and power supply system.

To evaluate the integration between the hardware and software subsystems, a positioning experiment was performed using the twelve Dynamixel AX-12A actuators. Each motor was commanded to move through ten predefined angular positions under no-load conditions, performing 30 repetitions per position. The angular positions reported by the actuators were compared with the commanded references to assess the overall communication and control performance of the platform.

TABLE III
Communication error percentages of the Dynamixel AX-12A+ motors

Motor ID	Communication error (%)
1	0.56
2	0.56
3	1.94
4	0.56
5	1.67
6	0.28
7	0.28
8	0.83
9	1.39
10	1.11
11	1.11
12	0.83

The results are summarized in Table III. All actuators exhibited positioning errors below 2%, with an average error of approximately 0.93%. These results demonstrate reliable communication between the ROS 2 framework, the embedded

controller, and the actuator network, ensuring accurate transmission of control commands and feedback information.

B. Trajectory Performance of the Robotic Leg

Three trajectory configurations were analyzed. The first configuration used $a = 9$ cm and $h = 2$ cm, producing a wide horizontal displacement with limited vertical lifting. The second configuration used $a = 3$ cm and $h = 6$ cm, which produced an unstable behavior because the required motion exceeded the mechanical or kinematic limits of the leg. The third configuration used $a = 3$ cm and $h = 2$ cm, producing the most suitable response among the tested cases, since the generated trajectory remained within the feasible workspace of the mechanism. The evaluated configurations are presented in Fig. 3. In the figure, the blue, green, and red curves represent the hip, femur, and tibia joint trajectories, respectively.

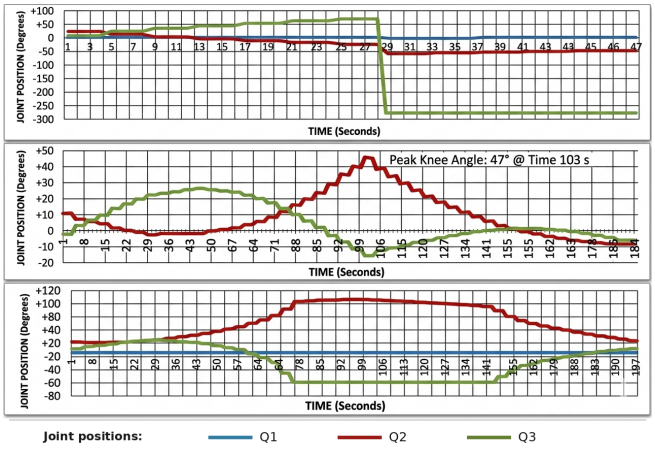


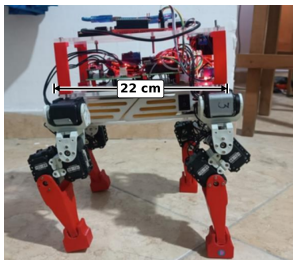
Fig. 3. Single-leg trajectory evaluation under different stride length (a) and foot clearance (h) parameters. The top trajectory corresponds to $a = 9$ cm and $h = 2$ cm, the middle trajectory to $a = 3$ cm and $h = 6$ cm, and the bottom trajectory to $a = 3$ cm and $h = 2$ cm.

The experimental evaluation showed that the semi-elliptical trajectory parameters directly affect the feasibility of the leg motion. Excessive vertical displacement may lead to unreachable configurations or abrupt joint behavior, while a moderate combination of horizontal and vertical motion improves the continuity of the generated trajectory.

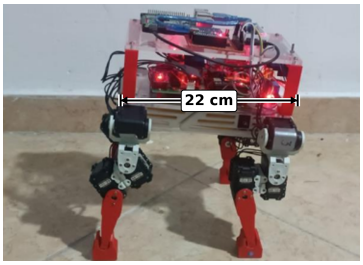
C. Evaluation of Robot Stability Under Roll and Pitch Variations

The stability of the proposed quadruped robot was evaluated under different static body orientations by modifying the roll and pitch angles of the platform. This experiment aimed to verify the capability of the kinematic model and control architecture to maintain stable postures while adapting the robot body to uneven terrain conditions. Positive and negative rotations were applied around both the longitudinal ((x)) and lateral ((y)) axes, and the resulting robot configurations were analyzed. The evaluated postures are illustrated in Fig. 4.

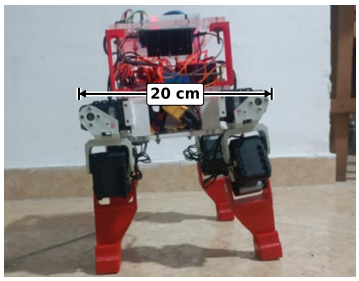
The results demonstrate that the proposed platform was able to maintain stable static configurations under both positive and negative roll and pitch variations. The inverse kinematic model successfully adjusted the leg positions to preserve ground contact and body balance while modifying the orientation of the robot.



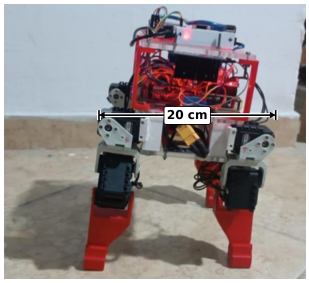
(a) Positive around the x-axis (roll angle).



(b) Negative around the x-axis (roll angle).



(c) Positive around the y-axis (pitch angle).



(d) Negative around the y-axis (pitch angle).

Fig. 4. Static stability evaluation under positive and negative roll and pitch orientations. (a) Positive roll, (b) negative roll, (c) positive pitch, and (d) negative pitch.

D. Locomotion Positioning Accuracy

The locomotion accuracy of the proposed quadruped robot was evaluated through a 50 cm linear displacement task. The robot was commanded to travel from a predefined starting position using the proposed gait generation and kinematic control framework, and the difference between the desired and actual displacement was recorded. A total of 30 trials were conducted under identical conditions. The experimental results yielded a mean positioning error of 4.53 cm, an RMSE of 5.62 cm, and a percentage error of 9.06%, demonstrating the feasibility of the proposed locomotion system for research and educational applications. The statistical indicators obtained from the experiments are summarized in Table IV.

TABLE IV
LOCOMOTION POSITIONING ACCURACY METRICS

Metric	Value
Target displacement (cm)	50.0
Mean displacement (cm)	45.3
Mean absolute error (cm)	4.53
RMSE (cm)	5.62
Standard deviation (cm)	3.28
Percentage error (%)	9.06

Although a deviation from the desired position was observed from the desired position, the robot successfully executed the

planned locomotion tasks using the proposed kinematic and trajectory generation framework. The observed error is mainly attributed to the open-loop control architecture, actuator positioning uncertainties, and mechanical tolerances. Nevertheless, the results validate the integration of the mechanical, electronic, communication, and software subsystems, demonstrating the feasibility of the proposed low-cost quadruped platform for educational and research applications.

V. Conclusion

This paper presented the design, implementation, and experimental validation of a low-cost quadruped robotic platform developed using commercially available components, additive manufacturing techniques, and open-source software tools. The proposed architecture integrates ROS 2 as the high-level robotic framework, CAN-based communication for distributed control, and Dynamixel actuators for locomotion. Experimental results demonstrated the successful operation of the platform, validating the integration of its mechanical, electronic, and software subsystems. Structural analyses supported the selection of suitable manufacturing materials, while communication and actuator tests confirmed reliable system performance. Furthermore, locomotion experiments showed that the robot was capable of executing the desired movements with a mean positioning error of 4.53 cm and a percentage error of 9.06% over a 50 cm displacement task, demonstrating the feasibility of the proposed locomotion framework.

The main contribution of this work is a low-cost, reproducible quadruped robotic platform with a total implementation cost of approximately USD 800. Commercially available components and additive manufacturing support its fabrication, while open-source design files, control software, and development documentation provide the resources needed for replication. This combination offers an accessible platform for robotics education and locomotion research in institutions with limited budgets, particularly in Latin America. Future work will focus on closed-loop locomotion control, inertial measurement units and vision-based perception, and improved gait generation and navigation algorithms to increase accuracy, robustness, and terrain adaptability.

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